

Omar Obaid Alkhammash

ML/AI Engineer - ML/AI Specialist - ML/AI Researcher - Data Analyst

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PROFILE

Artificial Intelligence graduate from the University of Jeddah with strong experience in Business Intelligence, data analytics, and predictive modeling. Selected among the **top 1% of 23,000+ participants** in the **KAUST Academy** AI program. Authored comprehensive Business Requirements Documents (BRD) and built multi-model dashboards for complex data visualization. Former **Data Analyst Intern at Makkah Health Cluster**, working with large-scale healthcare data (91,000+ records), SQL, and Power BI to generate actionable AI-driven insights.

EDUCATION

University of Jeddah <i>Bachelor's Degree, Artificial Intelligence (GPA: 4.68/5)</i>	Jeddah, Makkah, Saudi Arabia <i>Aug 2021 – Jul 2026</i>
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PROFESSIONAL EXPERIENCE

KAUST Academy AI Specialization Program Internship <i>KAUST (King Abdullah University of Science and Technology)</i> • Machine Learning Engineering: Engineered advanced AI solutions as part of the top 1% cohort (100 selected from 23,000+ applicants), including Computer Vision and NLP models. • Deployed 2 local LLM Agents (Qwen and Gemma) utilizing MCPs to automate complex agentic workflows. • Implemented end-to-end MLOps pipelines using Docker and MLflow to streamline model deployment.	June 2026 – Aug 2026 <i>Thuwal, Makkah, Saudi Arabia</i>
Data Analyst Intern <i>Makkah Health Cluster</i> • Analyzed 91,000+ raw patient records during the Hajj season, utilizing Power BI and SQL to develop 3 interactive healthcare reports. • Built AI-driven predictive models utilizing Hajj patient datasets to forecast future healthcare trends and system demands.	June 2025 – Aug 2025 <i>Makkah, Makkah, Saudi Arabia</i>

PROJECTS

Healthcare AI Business Requirements (BRD) <i>Project Lead Business Analysis & AI Strategy</i> • Authored a comprehensive Business Requirements Document (BRD) detailing real-time data ingestion, sub-second mobile alerting, and UI dashboards for a healthcare AI system. • Defined stakeholder and technical requirements for a Vision Transformer (ViT) model achieving 90% accuracy, ensuring 99.9% system uptime and zero privacy risk.
JEPA Text-To-Speech (Arabic & English) <i>Team Lead Deep Learning & Speech Synthesis</i> • Developed a novel bilingual Text-To-Speech AI model handling 2 languages (Arabic and English) utilizing the Joint-Embedding Predictive Architecture (JEPA).
Human Activity Recognition via Wi-Fi CSI <i>Team Lead Deep Learning & Signal Processing</i> • Engineered a full-stack AI application utilizing Wi-Fi signals to classify human activities without visual sensors, achieving 97% classification accuracy. • Spearheaded synthetic data generation techniques that doubled the database size from 4,000 to 8,000 Wi-Fi signal images, significantly enhancing model robustness.
AI Financial Fraud Sentinel <i>Team Lead Deep Learning & Cybersecurity</i> • Engineered a multi-model cybersecurity dashboard with Streamlit, utilizing Random Forest, PyTorch Autoencoders, and Isolation Forests to detect anomalous transactions. • Automated a data pipeline to process 6.3M+ records with MLflow, achieving a 99.45% fraud detection recall rate on highly imbalanced data.

TECHNICAL SKILLS

Domains & Competencies: Software Engineering, Data Analysis, Business Intelligence, Project Management
Programming: Python, C++, SQL, Java, JavaScript (Node.js), Rust, GO, Object Oriented Programming (OOP)
AI & Machine Learning: Deep Learning, NLP, Reinforcement Learning (RL), Computer Vision, Generative AI, TensorFlow, PyTorch, Scikit-learn
Data & Visualization: Data Science, Power BI, NumPy, Pandas, Matplotlib, Seaborn
Tools & Cloud: GCP, Azure, AWS, MLOps, MLflow, Docker, Anaconda/Miniconda, Prompt Engineering
Languages: Arabic (Native), English (Fluent - IELTS Academic: 7.5 STEP: 93)

COURSES

- The Advanced & Introductory AI Specialization Programs, KAUST Academy
- Deep Learning Specialization, DeepLearning.AI
- Advanced Pathways in Artificial Intelligence, SDAIA
- Microsoft Power BI Desktop for Business Intelligence, Maven Analytics